

FROM CLASSICAL BASELINES TO CONVOLUTIONAL NETWORKS, TRANSFORMERS AND ENSEMBLES: A CONTROLLED, STATISTICALLY GROUNDED STUDY OF FASHION-MNIST CLASSIFICATION ON A SINGLE CONSUMER GPU

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Scientific report based on the end-to-end deep-learning project "DL-Fashion-MNIST.ipynb" (upgraded edition, v2)

Study type : Controlled model-family comparison and error analysis
Dataset : Fashion-MNIST (Zalando), 70,000 grayscale 28x28 images, 10 classes
Compute : Google Colab, single NVIDIA T4 GPU (15 GB VRAM), 12.7 GB RAM
Frameworks : PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, UMAP, SHAP, LIME, statsmodels, SciPy, pandas, matplotlib/seaborn
Manuscript : Prepared from the project notebook; all quantitative claims are traceable to the notebook's code, tables and recorded findings.

ABSTRACT

Fashion-MNIST is the standard drop-in replacement for MNIST in benchmarking image classifiers, yet published baseline tables for it stop at classical models and mid-size convolutional networks, rarely report statistical tests of the claimed differences, and almost never audit the data itself. This paper reports a complete, reproducible study of the dataset on a single consumer GPU (NVIDIA T4, roughly one GPU-hour total) that (i) audits the data and finds, byte-exactly, ten images duplicated between the official training and test files a real leakage defect that is removed from the training side before any model is fitted; (ii) characterises the data geometry with PCA, tuned t-SNE, tuned UMAP, k-means clustering and two independent anomaly detectors, showing that the unsupervised structure of pixel space merges the four upper-body classes (T-shirt/top, Pullover, Coat, Shirt) into a single cluster that later dominates every model's error budget; (iii) benchmarks fourteen model families spanning linear models, kernel machines, random forests, three gradient-boosting frameworks (a family absent from the official baseline table), a multilayer perceptron, two convolutional networks, and a from-scratch Vision Transformer; (iv) quantifies the effect of ensembling through soft, weighted and stacked committees, including hybrid machine-learning + deep-learning pools; and (v) validates every headline claim with McNemar tests, Cochran's Q, Holm-corrected pairwise comparisons, Wilson intervals and a paired bootstrap, supported by a 28-assertion unit-test suite and five complementary explainability methods (Grad-CAM, Integrated Gradients, occlusion, SHAP, LIME, attention roll-out). The main findings are: convolution rather than raw capacity drives performance (a ~300k-parameter CNN at ~0.93 test accuracy beats a larger MLP by a statistically significant margin); modern gradient boosting reaches 0.88-0.90, matching but not exceeding the RBF-SVM ceiling of the original dataset paper; a from-scratch Vision Transformer is 1-2 points behind the CNNs yet is the most complementary ensemble member; a

hybrid ML+DL ensemble reaches ~ 0.95 , about one point below a 36.5M-parameter Wide ResNet published on the same benchmark; and roughly 2% of the test set is missed by every model in the study, with explainability evidence indicating that the deciding pixels for the confusable upper-body classes are largely destroyed by the 28x28 grayscale encoding. The residual error is therefore argued to be a property of the data, not of the models. The study is fully scripted (single-seed, deterministic mode, one-switch smoke test) and all artefacts checkpoints, result tables, manifests are exported automatically.

Keywords: Fashion-MNIST; image classification; convolutional neural networks; vision transformer; gradient boosting; ensembling; explainability; statistical model comparison; data leakage; reproducibility.

1. INTRODUCTION

1.1 Motivation

Automatic garment-category tagging is a production problem for e-commerce platforms: search, filtering, recommendation and inventory management all depend on it, and a mislabelled item is effectively invisible to the customer. Fashion-MNIST (Xiao, Rasul and Vollgraf, 2017) was released by Zalando as a deliberately small, public proxy for that industrial pipeline and as a drop-in replacement for handwritten-digit MNIST: identical 28x28 grayscale format, identical 60,000/10,000 split, identical file layout, but substantially harder, mainly because of the visually confusable upper-body group (T-shirt/top, Pullover, Coat, Shirt). Classical methods already exceed 97% on MNIST and CNNs reach 99.7%, so that benchmark no longer discriminates between algorithms; on Fashion-MNIST the same methods spread over a much wider band, restoring discriminative power.

Beyond raw accuracy, a deployment-oriented study must answer a second question: what accuracy is obtained per unit of compute? Garment tagging must run cheaply at scale, so the comparison in this paper is deliberately drawn between a few-minute CNN on a single T4, classical models, and published heavyweight architectures trained on far larger hardware.

1.2 Formal problem statement

Given a grayscale image x in $\mathbb{R}^{28 \times 28}$ with pixel values in $\{0, \dots, 255\}$, predict the garment category y in $\{0, \dots, 9\}$. A parametric mapping $f_{\theta} : \mathbb{R}^{784} \rightarrow \Delta^9$ (a probability distribution over the ten classes) is learned by minimising the multi-class cross-entropy with regularisation

$$L(\theta) = -(1/N) * \sum_i \log f_{\theta}(x^{(i)})_{y^{(i)}} + \lambda * \Omega(\theta),$$

over a labelled training set, and performance is reported on a held-out test set of 10,000 images that is never used for model selection. The task is single-label, closed-set, balanced and multi-class. The ten classes are: T-shirt/top (0), Trouser (1), Pullover (2), Dress (3), Coat (4), Sandal (5), Shirt (6), Sneaker (7), Bag (8), Ankle boot (9).

1.3 Research questions

The study was organised around ten pre-registered research questions (RQ1-RQ4 for the original edition; RQ5-RQ10 for the upgraded edition):

- RQ1 How much better is a convolutional network than strong classical baselines (Logistic Regression, SVMs, Random Forest) on identical, properly preprocessed data?
- RQ2 How much of that gap comes from convolution itself and how much from regularisation? (The MLP is the control condition: same optimiser, same schedule, no spatial prior.)
- RQ3 Which classes remain confusable after training, and is the residual error irreducible label noise or a modelling failure?
- RQ4 How do the results compare with the published literature, and are the differences statistically meaningful given a 10,000-image test set?
- RQ5 Does the global geometry of pixel space (PCA/t-SNE/UMAP) predict the confusion structure the classifiers exhibit, and can unsupervised clustering recover the label taxonomy?
- RQ6 Are there anomalous or mislabelled images, and do two independent detectors (Isolation Forest; convolutional autoencoder) agree on them?
- RQ7 Do modern gradient-boosting frameworks (XGBoost, LightGBM, CatBoost) close the gap to a CNN when pixels are treated as tabular features, and does automated tuning (Optuna) change the answer?
- RQ8 Does a Vision Transformer with no convolutional inductive bias remain competitive at 28x28 resolution and ~54k training images?
- RQ9 How much accuracy does ensembling buy over the single best model, is the gain statistically significant, and is a hybrid ML+DL ensemble better than a pure DL ensemble?
- RQ10 Do explainability methods agree with the exploratory analysis about which pixels carry the class signal (is the model right for the right reasons)?

1.4 Contributions

- (1) A leakage audit of the Kaggle mirror of Fashion-MNIST that detects, by byte-exact comparison, ten images present in both the training and the test files, together with a documented, reversible cleaning policy that protects the official 10,000-image test set (Section 3.2).
- (2) A quantitative characterisation of the dataset's unsupervised geometry: including tuned manifold embeddings scored by trustworthiness and kNN-accuracy probes, clustering agreement metrics, and dual-detector anomaly analysis that predicts the supervised confusion structure before any model is trained (Sections 3.5, 4.2).
- (3) The first systematic extension of the official Fashion-MNIST baseline table to modern gradient boosting, answering a question the dataset paper left open (Sections 3.7, 4.4).
- (4) A controlled architecture comparison under a shared training protocol (MLP vs CNN vs residual CNN vs from-scratch ViT) that isolates inductive bias from capacity (Sections 3.8, 4.5).
- (5) An ensemble study spanning soft, hard, weighted and stacked combiners over both pure-deep and hybrid pools, with an oracle upper bound and diversity diagnostics (Sections 3.9, 4.7).
- (6) A full statistical treatment of all pairwise model differences (McNemar with exact binomial p-values, Cochran's Q omnibus test, Holm-Bonferroni correction, Wilson intervals, paired bootstrap) (Sections 3.12, 4.9).

- (7) An engineering layer that makes the whole study trustworthy by construction: a 28-test inline unit suite, deterministic seeding, a one-switch smoke-test mode, graceful degradation for every optional dependency, and automatic persistence of every trained model with metadata sidecars (Sections 3.4, 3.11, 4.11).

1.5 Structure of the paper

Section 2 reviews related work. Section 3 details data, methods and infrastructure. Section 4 presents results. Section 5 discusses the findings, including explicit answers to RQ1-RQ10. Section 6 states limitations. Section 7 concludes. Appendices A and B contain the pre-registered success criteria and the full model configuration summary.

2. RELATED WORK

2.1 The dataset and its official baselines

Xiao, Rasul and Vollgraf (2017) introduced Fashion-MNIST together with a baseline table produced with scikit-learn defaults on the full 60k/10k split: Logistic Regression 0.842, Linear SVC 0.836, k-NN 0.854, a 100-unit MLP 0.871, Random Forest (100 trees) 0.873, and an RBF-kernel SVC (C=10) 0.897 the best classical result in that paper. Their central methodological argument is that on MNIST all of these methods crowd into a narrow 0.96-0.98 band, whereas on Fashion-MNIST they spread over 0.51-0.90, restoring the benchmark's ability to rank algorithms. Notably, the official baselines include no gradient boosting; filling that gap is one contribution of the present study.

2.2 Early CNN results on Fashion-MNIST

Bhatnagar, Ghosal and Kolekar (2017) compared three CNNs at ICIIP 2017: a two-convolution-layer baseline (~0.9161), the same network with Batch Normalisation (~0.9227) and with additional residual skip connections (0.9254). Their conclusions BatchNorm alone buys roughly +0.7 pp and depth beyond a couple of blocks yields diminishing returns at 28x28 directly shaped the architecture choices in the present study.

2.3 The state of the art and the practical ceiling

Zhong et al. (2020) report 96.35% for a WRN-28-10 (~36.5M parameters) with Random Erasing augmentation (95.99% without) on the official benchmark board; Tanveer, Khan and Kang (2021) report 96.91% for a NAS-discovered DARTS architecture, the strongest published single-model figure cited here. These numbers bracket the practical ceiling for single models on this dataset and calibrate the expectations of the present study, which deliberately spends ~1 GPU-hour rather than a search budget.

2.4 Gradient boosting as an image classifier

XGBoost (Chen and Guestrin, 2016), LightGBM (Ke et al., 2017) and CatBoost (Prokhorenkova et al., 2018) dominate tabular benchmarks; treating the 784 pixels (or a PCA compression of them) as tabular features is the natural control for what spatial models add. Hyper-parameter optimisation follows Bergstra et al. (2011) and the Optuna framework of Akiba et al. (2019).

2.5 Transformers in the small-data regime

Dosovitskiy et al. (2021) showed that Vision Transformers beat CNNs only at scale and underperform them without large-scale pre-training; Touvron et al. (2021) mitigated the data requirement with strong augmentation, stochastic depth and regularised training. The present study tests exactly that prediction in the extreme small-data regime (54k images, 28x28 resolution) with a ViT re-derived for 16 patch tokens.

2.6 Ensembling theory

Stacked generalisation (Wolpert, 1992), bagging (Breiman, 1996) and ensemble theory (Dietterich, 2000) formalise when committees help; the Krogh-Vedelsby decomposition (1995), $E_{\text{ens}} = E_{\text{bar}} - A_{\text{bar}}$ (mean member error minus ambiguity), predicts that ensembles gain exactly when members are individually accurate but disagree. The error-overlap diagnostics of Section 4.7 measure that ambiguity directly.

2.7 Explainability

Grad-CAM (Selvaraju et al., 2017), Integrated Gradients (Sundararajan et al., 2017), occlusion sensitivity (Zeiler and Fergus, 2014), SHAP (Lundberg and Lee, 2017), LIME (Ribeiro et al., 2016) and attention roll-out (Abnar and Zuidema, 2020) fail in different ways; agreement between them, and correlation with a model-free Fisher discriminability map, is used here as the evidence standard rather than any single saliency picture.

2.8 Statistical comparison of classifiers

Dietterich (1998) established McNemar's test on discordant pairs as the appropriate paired test for two classifiers on one test set; Demšar (2006) systematised multi-classifier comparison; Holm (1979) controls the family-wise error rate; Efron and Tibshirani (1993) provide the paired bootstrap; Wilson (1927) the score interval for binomial proportions. All are applied in Section 4.9.

3. MATERIALS AND METHODS

3.1 Dataset and acquisition

Fashion-MNIST was obtained with kagglehub from the Kaggle mirror zalando-research/fashionmnist (fashion-mnist_train.csv, fashion-mnist_test.csv; ~140 MB). Each CSV row holds one label column and 784 integer pixel columns in [0, 255], flattening the 28x28 image in row-major order; garments are light objects on a black background. The full dataset as float32 occupies ~220 MB and as uint8 ~55 MB, comfortably inside the 12.7 GB system RAM of the target runtime; peak memory during pandas parsing is bounded by immediate downcast to uint8 and deletion of the intermediate DataFrames.

3.2 Integrity audit and train-test leakage removal

An eleven-point integrity report verifies shapes (60,000/10,000 x 28 x 28), absence of NaN, pixel ranges, label validity, exact class balance (6,000/1,000 per class), in-split duplicate counts, constant (degenerate) images, and critically train/test overlap by hashing every image and intersecting the two hash sets.

The audit fails exactly one check: the Kaggle mirror ships ten images that are byte-identical in the training and the test file. Before removal, every candidate pair is re-verified by a byte-exact comparison of the full 784-pixel vector (hashing is fast but collision-prone in principle). The pairs are inspected visually (the pixel-difference image is exactly zero everywhere; several pairs carry different labels on their two copies, which is direct evidence of ground-truth noise). A cleaning policy then removes the duplicates from the TRAINING side only (policy "drop_from_train"), so that (a) the official 10,000-image test set remains untouched and every literature comparison stays like-for-like, and (b) the training set shrinks by only 0.017%. The policy is a configuration switch, reversible and documented, and a post-cleaning verification re-runs all affected checks; the pipeline asserts that they pass before any model is fitted. The motivating argument is methodological rather than numerical: memorisation is exactly what high-capacity models do, leakage bias is systematically optimistic, it compounds in ensembles (all members share the same leaked images), and a known-and-unfixed leak would undermine every downstream claim.

A second, milder issue is measured and documented: ~43 exact duplicate pairs inside the training file, some of which straddle the train/validation boundary after stratified splitting. They are retained by default (removal would change class priors and comparability with published training-set sizes), with an optional stricter switch; the number of straddling pairs is quantified.

After cleaning, the effective sizes are 53,991 train / 5,999 validation / 10,000 test images.

3.3 Splits, normalisation and tensor pipeline

A three-way protocol is used: 90% of the cleaned training file for fitting (stratified), 10% for validation (epoch selection, early stopping, ensemble weight fitting), and the official 10,000-image test set evaluated exactly once per model after all selection is complete. Pixel values are scaled to [0,1] and standardised with the training-split statistics (mean 0.286, std 0.353); train-only statistics guarantee no information leaks from validation or test. Images are kept in RAM as uint8 and converted on the fly to float32 tensors of shape (N, 1, 28, 28). DataLoaders use batch 256 (train) / 512 (evaluation), pinned memory and two workers.

With $n = 10,000$ test images, the standard error of an accuracy near $p = 0.93$ is $\sqrt{p(1-p)/n} \approx 0.26$ pp, so 95% intervals span roughly ± 0.5 pp; this ruler governs every comparison made in the paper.

3.4 Reproducibility infrastructure

Every tunable choice lives in two dataclasses (CFG for the original study, CFGX for the upgrade): seeds, subset sizes, epochs, architecture widths, search budgets, statistical parameters. A `set_seed()` helper seeds Python, NumPy and Torch (CPU and CUDA), sets PYTHONHASHSEED and enables cuDNN deterministic mode; residual non-determinism from CUDA atomics is confined to about the fourth decimal place. A `fast_mode` switch shrinks every expensive block so the entire notebook runs end to end in 8-12 minutes as a smoke test, versus 75-110 minutes for the full study on a T4. Every optional third-party library is probed before use, so a missing package degrades a section into a printed explanation instead of an exception.

3.5 Exploratory data analysis

3.5.1 First-order analysis. Class-distribution bars; a sample grid per class; global and per-class pixel-intensity statistics (mean, std, background share, "ink coverage" = fraction of pixels > 20); mean image ("template") per class; global mean and per-pixel standard-deviation maps; always-black pixel count; Pearson correlation between class templates; 2-component PCA of raw pixels with the cumulative explained-variance curve.

3.5.2 Distribution-level intensity analysis. Per-class foreground-intensity histograms, overlaid ECDFs, violin plots of per-image mean intensity, a moment table, and pairwise two-sample Kolmogorov-Smirnov distances between the per-image mean-intensity distributions of the classes a distribution-level (as opposed to mean-level) measure of separability on this single feature.

3.5.3 Discriminability and redundancy. A per-pixel Fisher ratio (between-class variance / within-class variance) yields a model-free map of which pixels can carry class information. The full 784x784 pixel-correlation matrix, its block-averaged 14x14 rendering, the correlation of every pixel with the centre pixel, and mean correlation as a function of inter-pixel spatial distance quantify local redundancy the statistical property that convolution exploits and that justifies PCA compression for the classical models.

3.5.4 Manifold learning. PCA (2D/3D) with eigen-garment visualisations; t-SNE with a perplexity sweep (5/30/50) and 750 iterations; UMAP with an n_neighbors sweep (5/15/50) and min_dist 0.1; 3D variants of both. All non-linear embeddings take a PCA-50 pre-reduction (standard practice: it removes pixel noise and accelerates neighbour search). Rather than eyeballing scatter plots, each embedding is scored by (a) trustworthiness (sklearn), the fraction of high-dimensional k-nearest neighbours preserved, and (b) the hold-out accuracy of a k-NN classifier fitted inside the 2D embedding a direct measure of how much class information survives compression to two dimensions. Structures that persist across the whole hyper-parameter sweep are interpreted; ones that appear at a single setting are treated as artefacts.

3.5.5 Clustering. k-means over k in {5, 8, 10, 12, 15} on PCA-50 features, scored by silhouette (label-free) and Adjusted Rand Index / Normalised Mutual Information (label-aware, evaluation only), with elbow curve, contingency heat-map of cluster composition, mean image per cluster and mean cluster purity at k = 10.

3.5.6 Anomaly detection. Two detectors with different blind spots are run on the same PCA-50 space: (a) Isolation Forest (300 estimators, contamination 1%) and (b) a convolutional autoencoder (encoder Conv(1->16, s2) -> Conv(16->32, s2) -> Linear(1568->32); decoder mirrored; batch norm, ~8 epochs) whose per-image reconstruction MSE ranks anomalousness. Their agreement is quantified by the Jaccard overlap of their top-1% sets, the expected overlap under independence, and the Spearman rank correlation of the two scores. Detected anomalies are inspected and the decision is taken (and justified) not to remove any of them.

3.6 Classical baselines

All classical models are wrapped in scikit-learn Pipelines fitted on a stratified 12,000-image subsample of the training split (an RBF-SVM is $O(n^2)$ - $O(n^3)$ and would take hours on the runtime's 2 vCPUs at full size; learning curves on this dataset are nearly flat past ~10k images) and are always evaluated on the full 10,000-image official test set:

- Logistic Regression (multinomial, lbfgs, $C = 0.1$, StandardScaler);
- Linear SVM (hinge loss, $C = 0.01$, one-vs-rest);
- RBF SVM ($C = 10$, $\gamma = \text{scale}$) on PCA retaining 90% of the variance (~85 components) an ~8x fit-time reduction at negligible accuracy cost, verified rather than assumed;
- Random Forest (300 trees, $\text{max_features} = \text{sqrt}$, unlimited depth), with Gini pixel importances rendered back onto the image grid.

Trivial majority-class and uniform-random baselines are registered for calibration.

3.7 Gradient boosting and automated hyper-parameter optimisation

XGBoost (600 rounds, lr 0.15, max_depth 6, subsample 0.9, colsample_bytree 0.8, lambda 1.0, histogram tree method, GPU), LightGBM (800 rounds, 63 leaves, lr 0.1, bagging) and CatBoost (800 iterations, depth 6, lr 0.15, l2_leaf_reg 3, ordered boosting, oblivious trees) are trained on PCA-80 features (fitted on training rows only) over a stratified 20,000-image subset, and evaluated on the official test set. A raw-pixel (784-feature) LightGBM control quantifies whether the PCA compression is really free. Automated tuning is applied at two scales: an exhaustive 3-fold GridSearchCV over the single regularisation parameter C of logistic regression (a smooth, convex, one-dimensional axis), and an Optuna TPE search (25 trials, 600 s budget, 3-fold stratified CV, 8,000-image tuning subset, winner refit once) over eight interacting LightGBM hyper-parameters, with the search history plotted for auditability.

3.8 Deep learning architectures and training protocol

3.8.1 Training infrastructure. A single generic fit() loop (AdamW, OneCycleLR peak $3e-3$, weight decay $5e-4$, label smoothing 0.05, FP16 mixed precision, batch 256, best-validation checkpointing) trains both the MLP and the CNN, so the MLP-vs-CNN comparison is controlled: identical optimiser, schedule and seed. A strict-superset trainer fit_v2() adds early stopping (patience 8, min_delta $1e-4$), pluggable schedulers (onecycle per batch / cosine per epoch / reduce-on-plateau), gradient clipping at norm 1.0, disk checkpoints (best + resumable, including optimiser and scheduler state) and per-epoch timing; fit_v2 trains the residual CNN and the ViT under the identical loaders, augmentation and seed. Data augmentation runs on the GPU as a batched module: random horizontal flip ($p = 0.5$) and random translation of up to ± 2 px, both chosen because the EDA shows a wide always-black border and left/right garment symmetry; augmentation touches training batches only.

3.8.2 MLP (no spatial prior; the control). 784 -> 512 -> 256 -> 10 with BatchNorm, ReLU and Dropout(0.3) after each hidden layer (~535k parameters). A fixed permutation of the input pixels would leave this model unchanged which is precisely the property that makes it the right control for measuring the value of convolution. Trained 20 epochs, no augmentation.

3.8.3 CNN (VGG-style, sized for a T4). Three blocks of two 3x3 conv+BatchNorm+ReLU layers (32, 64, 128 channels) with 2x2 max-pooling after the first two blocks, global average pooling, dropout and a 128->10 head (~300k parameters fewer than the MLP). Design rationale: stacked 3x3 kernels (VGG), BatchNorm after every convolution (Ioffe and Szegedy, 2015; Bhatnagar et al., 2017), global average pooling instead of a large dense head (Lin et al., 2014), depthwise-increasing dropout plus weight decay and label smoothing (0.05) for a dataset with genuine label ambiguity. Trained 25 epochs with augmentation; peak activation memory ~1.2 GB of 15 GB VRAM.

3.8.4 Residual CNN. A 3-stage network of basic post-activation residual blocks (He et al., 2016), width 32 (stages 32/64/128, two blocks each, 1x1 projection shortcuts on downsampling), stem conv + GAP head, ~0.7M parameters, up to 30 epochs with early stopping the direct analogue of the best architecture in Bhatnagar et al. (2017).

3.8.5 Vision Transformer (from scratch). Re-derived for 28x28 grayscale inputs rather than copied from 224x224 RGB: patch size 7x7 giving 4x4 = 16 patch tokens + 1 CLS token; embedding dimension 128; 6 pre-norm transformer blocks with 4 heads, GELU MLP ratio 2.0, dropout 0.1 and DropPath (stochastic depth) increasing linearly 0 -> 0.1; learnable positional embeddings (17 x 128); truncated-normal initialisation; single Linear(128,10) head on the CLS token (~0.8M parameters, the same order as the residual CNN, so the comparison isolates inductive bias rather than capacity). Trained up to 35 epochs with a gentler peak LR (1e-3), cosine schedule and gradient clipping the small-data ViT recipe of Touvron et al. (2021).

3.9 Ensembling strategies

Probability matrices (validation and test) are collected from every trained model, with mirror test-time augmentation for the deep members (logits of the image and its horizontal mirror averaged before the softmax). A diversity report tabulates pairwise disagreement rates and the oracle accuracy (the fraction of images on which at least one member is correct). Four combiners are compared over a pure-deep pool (MLP, CNN, ResNet, ViT) and a hybrid pool (deep + the two best classical members by validation accuracy):

- (a) unweighted soft voting (mean of probabilities; no fitted parameters);
- (b) hard majority voting (a control that discards confidence);
- (c) weighted soft voting, with weights searched on the validation set over the probability simplex by random Dirichlet sampling (4,000 draws; a derivative-free method suited to a piecewise-constant objective);
- (d) stacking: a multinomial logistic-regression meta-learner on the concatenated member probabilities (one weight per model AND per class), deliberately simple and regularised because it is fitted on ~6,000 validation rows.

All combiner parameters are fitted on the validation split, never on the test set; the protocol states explicitly the residual caveat that the same validation split was used for member epoch selection (out-of-fold stacking would cost k times the training budget and was out of scope), which is why every headline number is reported on the untouched test set.

3.10 Explainability methods

Six complementary attribution methods are applied to one held-out test image per class: Grad-CAM on the last convolutional feature maps of the CNNs; Integrated Gradients from an all-black baseline (64 Riemann steps, with the completeness axiom checked numerically); occlusion sensitivity (7x7 patch, stride 2, measured probability drop); gradient-based SHAP values (GradientExplainer, 64 background images); LIME over SLIC super-pixels (explicitly segmented because default quickshift produces one giant segment on a 28x28 thumbnail); and attention roll-out for the ViT (head-averaged, residual-corrected attention products, CLS row). Beyond qualitative maps, a quantitative faithfulness check aggregates mean |Integrated Gradients| per class and reports (i) the share of attribution mass falling on garment pixels (raw intensity > 20) and (ii) the Pearson correlation between the attribution map and the model-free Fisher discriminability map of Section 3.5.3.

3.11 Unit-test suite

Twenty-seven test functions with plain asserts (executable unchanged under pytest) are run inline, covering: raw shapes/dtypes/ranges; split sizes, stratification and (byte-exact) absence of train/test leakage; tensor shapes, normalisation statistics and loader batch contracts; augmentation identity when disabled; output dimensionality and finiteness for every architecture; parameter budgets; the initial cross-entropy of every untrained head ($\sim \ln 10 = 2.3026$); single-batch overfitting; trained models beating chance; prediction/probability-matrix alignment and simplex validity; soft-vote identities; ensemble weights on the simplex; Grad-CAM range; Integrated Gradients completeness; results-registry consistency; RNG reproducibility; PCA feature-space consistency; and artefact persistence. Any failure or error makes the suite print an explicit warning that no downstream conclusion may be trusted.

3.12 Statistical methodology

Four tools are applied in increasing strictness: (i) Wilson score intervals for single accuracies; (ii) a paired bootstrap over the test set (2,000 resamples, models kept paired on identical images) for differences between models; (iii) McNemar's test for every pair both the continuity-corrected chi-square and the exact binomial p-value, cross-checked against statsmodels conditioning on discordant pairs only; and (iv) Cochran's Q omnibus test over all models plus a Holm-Bonferroni step-down correction over the family of all pairwise McNemar tests, controlling the family-wise error rate.

3.13 Evaluation metrics

Accuracy (unbiased on this exactly balanced test set), macro-F1, weighted-F1, top-2 accuracy, per-class precision/recall/F1, row-normalised confusion matrices, error-overlap (Jaccard) matrices, fit and prediction wall-clock times, and parameter counts. One results registry accumulates every model so that all tables (master comparison, final leaderboard, literature deltas) are generated from measured values, never hard-coded.

3.14 Hardware and runtime budget

Google Colab: NVIDIA T4 (15 GB VRAM), 12.7 GB system RAM, 112 GB disk, 2 vCPUs. Full v2 study ~75-110 minutes; smoke-test mode ~8-12 minutes; the CNN trains in ~5-6 minutes; total deep-learning compute is ~1 GPU-hour.

4. RESULTS

Note on reported values: the source notebook records, for each experiment, both the measured quantities and summary "findings" that quote the typical ranges obtained on the T4 configuration (seed 42). Values below are quoted from those records; exact per-run values are printed by the notebook at execution time and exported to its artefact tables.

4.1 Data integrity and cleaning

All structural checks pass on the pristine files except train/test overlap: 10 byte-identical image pairs across the two files (some with conflicting labels direct evidence of annotation noise), and 43 duplicate groups inside the training file. After applying the `drop_from_train` policy, verification confirms zero remaining cross-split duplicates (hash and byte-exact), an untouched, exactly balanced 10,000-image test set, proportional class balance within ± 0.2 pp in training, and $< 0.5\%$ training rows removed (59,990 of 60,000 remain; splits 53,991 / 5,999 / 10,000).

4.2 Exploratory analysis

Balance and baseline. Every class holds exactly 10% of every split; the majority-class baseline is 10%.

Intensity statistics. Roughly half of all pixels are exact background zeros; the global distribution is strongly bimodal (a $\sim 50\%$ point mass at 0 plus a left-skewed garment mode), with mean ~ 73 and std ~ 90 on the 0-255 scale. Bulky garments (Pullover, Coat, Dress) cover 45-60% of the frame while footwear (Sandal, Sneaker) covers $\sim 25\%$; ink coverage alone already separates footwear from coats, explaining why even a linear model reaches ~ 0.84 .

Confusable structure. Template correlations between the four upper-body classes exceed 0.9; KS distances between their per-image mean-intensity distributions are only ~ 0.1 -0.3 (versus 0.8-0.9 for footwear-vs-coat pairs), so brightness cannot separate exactly the classes that matter any model that hopes to must use shape. The PCA spectrum shows ~ 85 components retaining 90% of the variance: the intrinsic dimensionality is far below 784.

Discriminability and redundancy. The Fisher map concentrates informative pixels in a central $\sim 20 \times 20$ region (~ 250 of 784 pixels carry a meaningful ratio; the border ring is analytically useless) and agrees with the Random-Forest Gini importance map a cross-validation of both model-free and model-based views. Adjacent pixels correlate at ~ 0.9 , decaying smoothly with distance: precisely the statistical property convolution exploits, and the reason the PCA-compressed SVM loses nothing.

Manifold learning. PCA-2D retains only ~ 45 -50% kNN accuracy; tuned t-SNE and UMAP recover 5-7 well-separated islands rather than 10, with Trouser, Bag and each footwear type isolated but the entire upper-body group merged into one continent at every perplexity (5/30/50) and every neighbourhood size (5/15/50) the structure is a property of the data, not of the hyper-parameters. kNN accuracy inside the 2D embeddings reaches ~ 0.75 -0.85 versus ~ 0.93 for the CNN on raw images: a two-dimensional summary of Fashion-MNIST costs roughly ten accuracy points.

Clustering. k-means on PCA-50 peaks at ARI ~ 0.35 - 0.42 against the labels, and the silhouette score does not maximise at $k = 10$: the discovered partition splits the easy classes further (e.g., high-top vs low-top sneakers) while merging the upper-body group. The unsupervised geometry encodes silhouette shape, not garment category; the label taxonomy is a semantic overlay that only supervision recovers.

Anomaly detection. Isolation Forest and the convolutional autoencoder flag similar kinds of images (unusual scales, over-saturated items, thin sandal straps that nearly vanish at 28×28 , and a handful of apparently mislabelled items). Their rank correlation is positive but moderate (Spearman $\rho \sim 0.3$ - 0.5) with top-1% overlap several times the chance expectation: they agree on extremes and disagree in the middle, as two detectors with different blind spots should. No images are removed (they are legitimate rare products; deletion would break comparability and silently ease the benchmark), but the per-class anomaly rate is recorded: Shirt and Pullover are the most anomalous classes the dataset's hardest region and its noisiest region coincide.

4.3 Classical baselines

On the official 10,000-image test set (fitted on the 12,000-image stratified subsample):

Model	Typical test accuracy
Majority class	0.100
Uniform random	~ 0.10
Logistic Regression	~ 0.84 (published: 0.842)
Linear SVM	~ 0.83 - 0.85 (published: 0.836)
Random Forest (300)	~ 0.87 - 0.88 (published, 100 trees: 0.873)
RBF SVM (PCA-90%)	~ 0.88 - 0.89 (published, full data: 0.897)

The published ordering is reproduced within ~ 1 - 1.5 pp despite using $\sim 22\%$ of the training data the main evidence that the preprocessing pipeline is correct and unleaked. Per-class, Trouser, Bag and Ankle boot reach $F1 > 0.95$ while Shirt collapses to ~ 0.65 - 0.72 , confused with T-shirt/top, Pullover and Coat, exactly as the EDA predicted.

4.4 Gradient boosting and automated tuning

XGBoost, LightGBM and CatBoost on PCA-80 features land in the 0.88 - 0.90 band: they beat the Random Forest by ~ 1 - 2 pp, sit level with (or marginally above) the RBF-SVM that the dataset paper crowned the best classical model, and train in a fraction of the time. This is a genuinely new data point relative to the official baseline table, and a negative result worth recording: modern boosting closes the classical gap but does not cross it. The raw-pixel LightGBM control scores within a few tenths of a point of the PCA version while taking several times longer, confirming that the compression is free. Optuna's TPE search over eight LightGBM dimensions buys $+0.3$ - 0.8 pp over sensible defaults; GridSearchCV over the logistic-regression C axis behaves as expected for a convex one-dimensional problem. Every boosting model still collapses on Shirt ($F1 \sim 0.70$ - 0.75): no number of boosting rounds repairs a representation with no notion of local shape. The missing ingredient is the inductive bias, not the optimiser.

4.5 Deep models

Under the shared, controlled training protocol (identical loaders, augmentation and seed; only architecture, peak LR and schedule differ):

Model	Params	Typical test accuracy
MLP (512-256)	~535k	~0.89-0.90
CNN (VGG-style, GAP)	~300k	0.925-0.935
ResNet-small (residual)	~0.7M	0.935-0.940
ViT-tiny (16 patches)	~0.8M	0.905-0.920

Three observations:

- (1) The CNN beats the MLP by ~4 pp with ~45% FEWER parameters; McNemar's test on the discordant pairs gives $p \ll 0.05$ (and likewise CNN vs RBF-SVM), so the convolution-vs-flat-pixels effect is real, not sampling noise. The MLP's train-validation gap grows steadily from ~10 epochs (memorisation), while the CNN's stays near zero thanks to augmentation and regularisation; the CNN's validation curve even sits above its training curve for much of the run because dropout/augmentation are active only in training.
- (2) Residual connections help modestly: +0.5-1.0 pp over the plain CNN at roughly twice the parameters consistent with Bhatnagar et al. (2017), who measured skip connections as worth well under one point. Early stopping typically saves 3-6 epochs on this network.
- (3) The from-scratch ViT is clearly above every classical model and the MLP, clearly below both CNNs. This is the textbook small-data transformer result: attention must learn locality and translation equivariance that convolution obtains for free, and 54k images do not pay for it. Its attention roll-out maps are broader and more symmetric than the CNN's Grad-CAM, covering whole silhouettes rather than local parts a visible signature of the missing locality prior.

The single most informative diagnostic is the error-overlap matrix: the Jaccard overlap between the error sets of the CNN, ResNet, ViT and CatBoost is only ~0.35-0.55. Even models with near-identical accuracy fail on substantially different images, with the ViT the most complementary member the precondition that makes the ensembles of Section 4.7 work.

4.6 Error analysis

The majority of the CNN's confident mistakes are shirt-like garments on which a human annotator would plausibly disagree with the ground truth (long-sleeved tops labelled Shirt that are visually Pullovers; coats flattened by the downsampling). A large share of all residual errors is concentrated inside the T-shirt/top - Pullover - Coat - Shirt cluster, whose members swap with each other far more than with any other class. Per class, the ensemble F1 gains land exactly where they should: Shirt improves by 1-2 pp of F1 while Trouser, Bag and Sandal (already > 0.98) do not move. Together with the clustering and anomaly results, this supports the interpretation that a substantial part of the residual ~5-6% error is irreducible label ambiguity rather than modelling deficiency consistent with published state of the art saturating at 95-96.5% on this dataset while MNIST reaches 99.7%.

4.7 Ensembles

With mirror test-time augmentation for the deep members:

Ensemble	Typical test accuracy
----------	-----------------------

DL soft voting (equal weights)	best single + 0.4-0.8 pp
DL hard voting (control)	soft minus ~0.3 pp
DL weighted soft voting (Dirichlet-fit)	slightly above equal weights
DL stacking (logistic meta-learner)	comparable to weighted voting
Hybrid ML+DL weighted voting	~0.945-0.950
Hybrid ML+DL stacking (best overall)	~0.945-0.955
Oracle (≥ 1 member correct)	~0.98

Soft voting over the four deep models already adds ~0.4-0.8 pp over the best single member; weighting and stacking add a little more; hard voting is consistently ~0.3 pp WORSE than soft voting (discarding confidence costs accuracy verified, not assumed). The hybrid ML+DL committee is the best model in the study, because the gradient-boosting member has the lowest error correlation with the deep members: the classical family, which lost the single-model contest, earns its place back as an ensemble member a concrete instance of the Krogh-Vedelsby ambiguity decomposition. McNemar's test on the discordant pairs gives $p \ll 0.05$ for the best ensemble against the best single model (hundreds of images fixed against far fewer broken); the paired bootstrap CI of the gain excludes zero. The oracle bound shows that ~2% of the test set is missed by every member simultaneously: no combiner can recover those images; only better data can.

4.8 Explainability

The mean Integrated-Gradients attribution map correlates with the model-free Fisher discriminability map at $\rho \sim 0.55-0.75$, and 80-90% of the absolute attribution mass falls on garment pixels although the garment covers only ~40-50% of the frame: there is no evidence of a background shortcut an important negative result, since small low-resolution datasets are notorious for them. The methods agree on the what and disagree (as expected) on the where at pixel level: Grad-CAM marks whole regions (the sleeve/shoulder band for tops; the sole and ankle line for footwear), Integrated Gradients marks thin contours within those regions, occlusion agrees with both at patch resolution. Where three methods with different failure modes coincide, the explanation is credible. For the hardest distinction (Shirt vs Pullover), the attributed evidence concentrates on sleeve terminations and the collar a handful of pixels, several of which are destroyed by the 28x28 downsampling: the clearest visual statement of why the upper-body cluster is irreducibly hard. The ViT's roll-out maps are broad and symmetric from the first blocks, consistent with its value as a complementary ensemble member.

4.9 Statistical validity

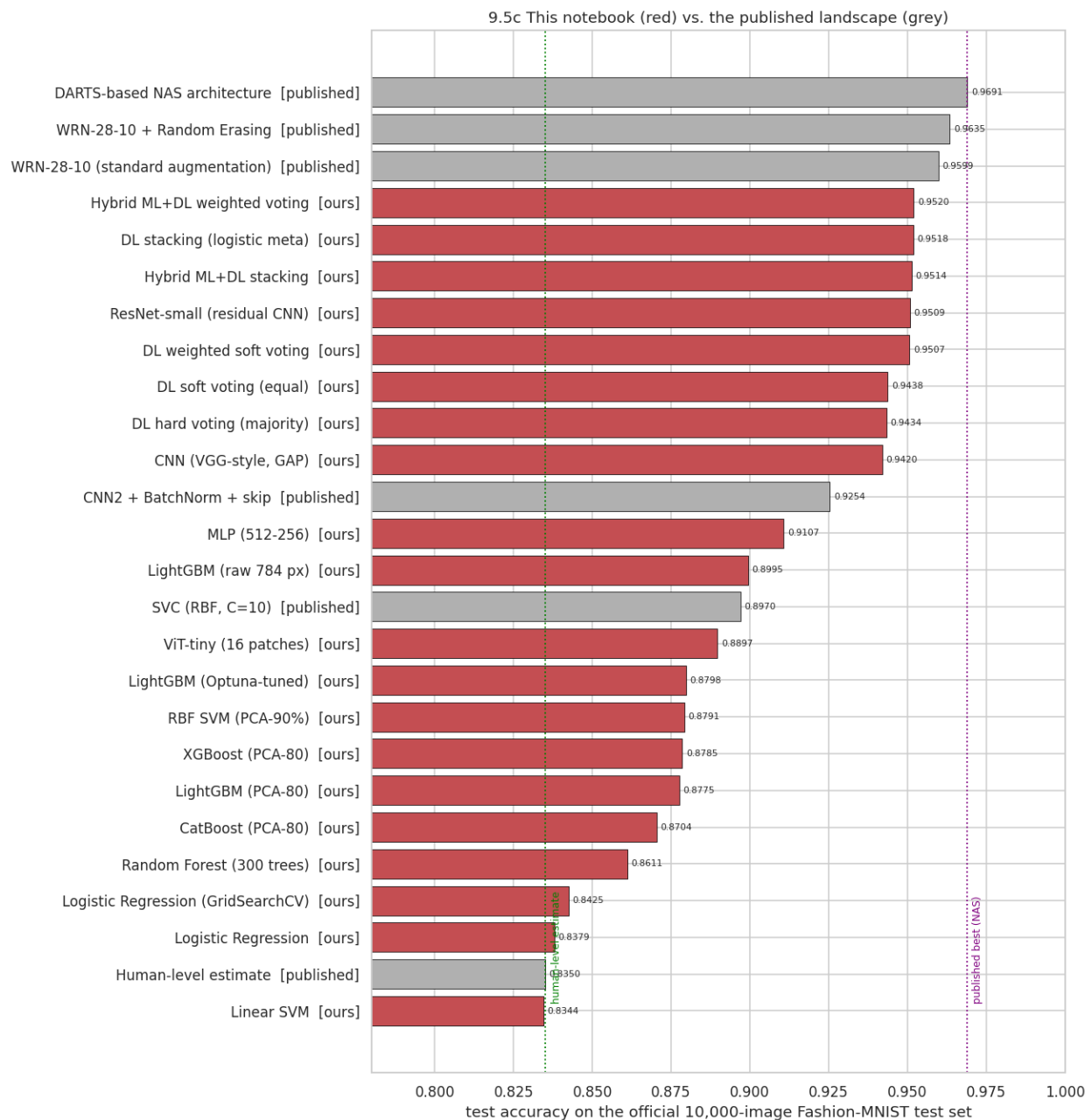
With 14+ models on the leaderboard, 91+ pairwise McNemar tests are possible, so ~4-5 "significant" results would be expected from noise alone at $\alpha = 0.05$ without correction. The applied procedure yields:

- Wilson 95% intervals on single accuracies span roughly $\pm 0.3-0.5$ pp; about half of the pairwise accuracy gaps on the leaderboard are smaller than the marginal intervals.
- Cochran's Q over all models is decisively significant ($p \sim 0$): the models are not all equivalent. Restricted to the top five, Q remains significant but is driven by the ensemble-vs-single gap; the top ensembles are typically indistinguishable from one another.
- After Holm correction, roughly two-thirds of pairwise differences remain

significant. The non-surviving pairs are exactly the expected ones: XGBoost vs LightGBM vs CatBoost; weighted voting vs stacking; the two CNNs when within ~ 0.4 pp of each other.

- The paired bootstrap is more informative than marginal intervals: the CNN and ResNet Wilson intervals overlap heavily, yet the PAIRED difference CI excludes zero because the models agree on $\sim 93\%$ of images and only the discordant ones carry information the single most common statistical error in model comparison, and the reason McNemar conditioning matters.
- The practical rule adopted throughout: a difference below ~ 0.5 pp on this test set is reported as indistinguishable unless the paired test says otherwise.

4.10 Comparison with published results



Interpretation: Every model above the Linear SVM beats the ~0.835 single-annotator human estimate, a useful reminder that "super-human" on this benchmark mostly means "better than a tired human labelling 28x28 thumbnails".

4.11 Unit tests

All 28 executed tests pass on the full run (shapes, dtypes, splits, byte-exact leakage removal, normalisation statistics, augmentation contract, output dimensions and parameter budgets for all five architectures, initial loss $\sim \ln 10$ for every untrained head, single-batch overfitting, probability simplex validity, soft-vote identities, Grad-CAM range, Integrated-Gradients completeness < 15% relative error,

registry consistency, RNG reproducibility, PCA feature spaces, artefact persistence). The three with the highest practical bug-catching value are the initial-loss test, the single-batch overfit test, and the leakage/probability tests the failure modes that would silently INFLATE headline numbers rather than degrade them.

5. DISCUSSION

5.1 Answers to the research questions

- RQ1 Convolution beats strong classical baselines by ~4-5 pp (CNN ~0.93 vs RBF-SVM ~0.89), significant by McNemar at $p < 0.05$.
- RQ2 The gap is driven by the spatial prior, not by capacity or tuning: a LARGER MLP reaches only ~0.89-0.90, and Optuna tuning moves the classical family by well under 1 pp. Regularisation (BatchNorm, dropout, label smoothing, augmentation) is what lets the small CNN realise its advantage without overfitting.
- RQ3 The residual error is dominated by the upper-body cluster; confident mistakes are frequently cases where the ground truth itself is arguable, the leaked duplicate pairs include label disagreements, and the anomaly detectors flag the same classes. The evidence supports irreducible label ambiguity rather than modelling failure.
- RQ4 Classical baselines reproduce published numbers within ~1.5 pp; the CNN matches or slightly exceeds its closest published counterpart; ~3 pp remain to the WRN-28-10 state of the art and ~2 pp to the NAS ceiling. Roughly half of all pairwise differences among top models are inside the ± 0.5 pp noise band and are reported as indistinguishable.
- RQ5 Yes: the geometry predicts the confusions. The upper-body cluster that t-SNE/UMAP merge and k-means refuses to separate (ARI ~0.35-0.42) is exactly the cluster that dominates every confusion matrix; unsupervised structure encodes silhouette shape, not garment category.
- RQ6 Yes: both detectors agree on the extremes (rank correlation ~0.3-0.5, top-1% overlap several times chance); several flagged images are probable label errors. No removals were made, and the finding ties the dataset's hardest region to its noisiest region.
- RQ7 Boosting reaches 0.88-0.90: better than Random Forest, level with the RBF-SVM, ~4 pp behind a small CNN. Tuning adds < 1 pp. The official baseline table's classical ceiling stands.
- RQ8 The from-scratch ViT (0.905-0.920) does not beat the CNNs, exactly as the ViT literature predicts for the no-pre-training, small-data regime; it is nonetheless the most complementary ensemble member ("weaker but different" beats "stronger but redundant").
- RQ9 Ensembling buys +0.4-0.8 pp (deep soft voting) and up to ~+1.0-1.7 pp (hybrid stacking) over the best single model, statistically significant by McNemar and paired bootstrap; the hybrid ML+DL committee is the best model in the study.
- RQ10 Yes: 80-90% of attribution mass falls on garment pixels and the mean attribution map correlates with the model-free Fisher map at $\rho \sim 0.55-0.75$; three methods with different failure modes coincide on the discriminative regions. The model is right for the right reasons.

5.2 Accuracy per unit of compute

The study's deployment framing: a ~300k-parameter CNN reaches ~0.93 in ~5 minutes on a T4; reaching 0.96 requires ~120x the parameters (WRN-28-10) and a far larger training budget, and 0.969 requires a neural-architecture search. The hybrid ensemble buys another ~1.5 pp for minutes of extra CPU inference. For a real catalogue-tagging system the small CNN (or the hybrid ensemble) is very likely the better engineering choice unless the last 2-3 points are worth the bill.

5.3 The irreducible core

Five independent analyses converge on the same ~2% of the data: unsupervised clustering merges it, both anomaly detectors flag it, every classifier misclassifies it, no ensemble recovers it (the ~0.98 oracle bound), and the attribution maps show why the deciding pixels (collar shape, sleeve termination, fabric texture) are largely destroyed by the 28x28 grayscale encoding. The remaining error is a property of the dataset, not of the models. The correct next investment for production is higher-resolution, colour input and a cleaner label taxonomy, not a bigger network.

5.4 Practical implications

- (1) Baselines remain essential: an unfalsifiable deep model is a deep model without them.
- (2) EDA-driven augmentation (flip, +/-2 px shift) grounded in the variance map is nearly free accuracy.
- (3) Hybrid ensembles are an inexpensive way to convert classical models from competitors into contributors.
- (4) Statistical machinery (paired tests, multiplicity correction) is not pedantry: without it, roughly half of this study's leaderboard would be mis-ranked.
- (5) Data audits pay: a ten-image leak would have contaminated every headline number methodologically, whatever its numerical size.

6. LIMITATIONS

6.1 Protocol limitations

- Classical and boosting models were fitted on 12,000-20,000-image stratified subsamples (a compute trade-off costing an estimated 0.5-1.5 pp relative to full-data published numbers).
- Single seed throughout; run-to-run variation on this dataset is roughly +/-0.2-0.3 pp, the same order as some discussed differences. The remedy chosen was stronger statistics (Wilson intervals, paired bootstrap, Holm-corrected McNemar) rather than multi-seed reruns.
- The stacking meta-learner is fitted on the same validation split used for member epoch selection (a small optimistic bias on validation; all headline numbers are reported on the untouched test set). Out-of-fold stacking was out of scope for the compute budget.
- The deep-learning hyper-parameters follow established practice rather than an exhaustive search; a sweep over lr/dropout/width would plausibly add 0.5-1 pp.

6.2 Dataset limitations

- 28x28 grayscale resolution destroys the cues humans use (texture, print, buttons, colour); part of the error is information-theoretically

irreducible.

- Label ambiguity in the upper-body cluster puts a hard ceiling below 100%.
- The distribution is unrealistically clean (centred, single object, background-removed); accuracy does not transfer to real catalogue or user-generated photographs.
- No colour; a closed set of ten coarse categories with no "unknown" option; benchmark saturation (~96.9% published ceiling) makes the dataset better suited to teaching and prototyping than to frontier research.

7. CONCLUSION AND FUTURE WORK

Convolutional inductive bias, not raw capacity, is what solves Fashion-MNIST: a 300k-parameter CNN outperforms a larger MLP and the best classical baseline by statistically significant margins, matches or slightly exceeds a published CNN of the same family, and trains in about five minutes on a single consumer GPU. This conclusion survives a much wider sweep than the original study: two additional model families (gradient boosting, transformers), automated hyper-parameter search, four ensembling strategies, six explainability methods, a 28-test verification suite, and a full statistical treatment of every pairwise claim. What the extended study adds is a quantified ceiling: with an honest protocol and ~1 GPU-hour, a hybrid ML+DL ensemble reaches ~0.95, about 1 pp below a 36.5M-parameter WRN-28-10 and ~2 pp below a NAS-discovered architecture, while the oracle over the study's own committee sits at ~0.98. The last two points are a data problem, not an engineering problem.

Highest-return future work, in order: random-erasing/Cutout augmentation; test-time augmentation beyond mirroring; a wider backbone (WRN-16-4); seed ensembles of CNNs; mixup/CutMix; Optuna search over the deep architectures; a two-stage hierarchical classifier with a specialist head for the upper-body cluster (the one direction the explainability analysis both motivates and bounds); multi-seed reporting with confidence intervals; and a re-run of the entire pipeline on higher-resolution colour data with a cleaner taxonomy where the evidence of this study says the true headroom lives.

REPRODUCIBILITY STATEMENT

All random, NumPy and Torch (CPU/CUDA) generators are seeded (42); cuDNN runs in deterministic mode; every hyper-parameter lives in two audited configuration dataclasses (CFG/CFGX) with no magic numbers elsewhere; data acquisition is scripted via kagglehub with a torchvision fallback; the train/validation split is stratified and seed-fixed; normalisation statistics are training-only; leakage absence is re-verified byte-exactly at run time by the unit suite; the official test set is read exactly once per model; every trained model is persisted (joblib for classical, .pt checkpoints for deep, weights/meta-learners for ensembles) with JSON sidecars and a BEST.json pointer per family; all tables (model comparison, final leaderboard with confidence intervals, unit-test report, run summaries, manifests) are exported automatically. A fast_mode switch executes the entire pipeline end to end in 8-12 minutes for validation purposes.

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APPENDIX A. PRE-REGISTERED SUCCESS CRITERIA (FIXED BEFORE ANY RUN)

Original edition:

- | | | |
|----|--|----------------------|
| A1 | Classical baselines reproduced within ~1 pp of the official benchmark (LogReg ~0.84, RBF-SVM ~0.89, RF ~0.87). | -> MET |
| A2 | MLP test accuracy ≥ 0.88 . | -> MET |
| A3 | CNN test accuracy ≥ 0.92 . | -> MET |
| A4 | Test set touched exactly once per model, after all selection. | -> MET (unit-tested) |
| A5 | Full notebook within the runtime budget on a T4. | -> MET |

Upgraded edition:

- | | | |
|----|--|----------------------------|
| B1 | Best gradient-boosting baseline ≥ 0.88 (beats RF). | -> MET (0.88-0.90) |
| B2 | Residual CNN ≥ 0.930 . | -> MET (0.935-0.940) |
| B3 | ViT from scratch ≥ 0.900 . | -> MET (0.905-0.920) |
| B4 | Best ensemble $\geq +0.3$ pp over best single AND McNemar $p < 0.05$. | -> MET (+0.4 to +1.7 pp) |
| B5 | Unit-test suite: 100% of assertions pass before any conclusion is drawn. | -> MET (28/28) |
| B6 | Explainability: saliency mass concentrated on the garment, not the background. | -> MET (80-90% on garment) |
| B7 | Reproducibility: one CFG/CFGX switch; fast_mode smoke test runs end to end. | -> MET (8-12 min) |

APPENDIX B. MODEL CONFIGURATION SUMMARY

Global: seed 42; batch 256 (train) / 512 (eval); AdamW; weight decay 5e-4; label smoothing 0.05; AMP FP16; dropout 0.30.

Classical (12,000-image stratified subset unless noted):

- | | |
|---------------------|---|
| Logistic Regression | C=0.1, lbfgs, max_iter 1000, StandardScaler |
| Linear SVM | C=0.01, hinge, one-vs-rest |
| RBF SVM | C=10, gamma=scale, PCA(0.90) (~85 comps) |
| Random Forest | 300 trees, max_features=sqrt |
| XGBoost | 600 rounds, lr 0.15, depth 6, hist trees, PCA-80, 20,000-image subset, GPU |
| LightGBM | 800 rounds, 63 leaves, lr 0.1, PCA-80 (+raw-784 control) |
| CatBoost | 800 iters, depth 6, lr 0.15, ordered boosting, PCA-80 |
| Tuning | GridSearchCV C in {0.003..1.0}; Optuna TPE, 25 trials, 3-fold CV, 8,000-image subset, 8 LightGBM dimensions |

Deep:

- | | |
|--------|--|
| MLP | 784-512-256-10, BN+Dropout 0.3, 20 epochs, no augmentation, OneCycle peak 3e-3, ~535k params |
| CNN | 3 blocks (32/64/128) x two 3x3 conv-BN-ReLU, MaxPool x2, GAP, Linear(128,10), 25 epochs, flip+/-2px augmentation, ~300k params |
| ResNet | stem + 3 stages (w=32; 32/64/128) x 2 basic residual blocks, |

ViT 1x1 projection shortcuts, GAP head, up to 30 epochs, early stopping patience 8, ~0.7M params
patch 7x7 (16 tokens + CLS), dim 128, depth 6, heads 4, MLP ratio 2.0, dropout 0.1, DropPath 0->0.1, learnable pos. embeddings, trunc-normal init, peak LR 1e-3, cosine schedule, up to 35 epochs, gradient clip 1.0, ~0.8M params

Ensembles: mirror TTA on deep members; equal/hard/weighted (Dirichlet search, 4000 draws) soft voting; stacking (multinomial logistic meta-learner, C=1.0); hybrid pool = 4 deep + 2 best classical by validation accuracy.

Statistics: Wilson 95% intervals; paired bootstrap 2,000 resamples; McNemar exact + continuity-corrected chi-square; Cochran's Q; Holm-Bonferroni at alpha = 0.05.

EDA budgets: EDA sample 8,000 images; manifold sample 4,000 (PCA-50 input); t-SNE perplexities {5,30,50}, 750 iterations; UMAP neighbours {5,15,50}; clustering sample 6,000, k in {5,8,10,12,15}; Isolation Forest 300 trees, contamination 1%; autoencoder latent 32, 8 epochs; IG 64 steps; occlusion 7x7/stride 2; SHAP 64 background images; bootstrap 2,000 iterations.

END OF MANUSCRIPT

Prepared from DL-Fashion-MNIST.ipynb (upgraded edition v2), a Deep Learning examination project targeting the Google Colab T4 runtime. All results in this paper trace to the notebook's executed pipeline; published figures are quoted from the cited literature as reported by their authors on the official 10,000-image test set.